

Round Table on “Freight Transport: Decarbonizing Maritime Shipping”

Shushma Sadanand, Group 01

Organized by: Audencia Business School, as part of the M1 Gaïa program.

Topic: "Decarbonizing Freight Transport: Rethinking Global Maritime Logistics for a Sustainable Future."

The round table discussion focused on: Re-evaluating the current model of global freight transport, particularly maritime shipping, which significantly contributes to greenhouse gas emissions. The panel explored how we can reduce the volume of goods being transported, adopt cleaner energy alternatives, and consider the true environmental cost of global logistics systems. The conversation highlighted the technical feasibility of decarbonizing freight while addressing political, economic, and social challenges.

This topic is especially relevant to young professionals and future business leaders because: It challenges conventional supply chain practices and encourages future leaders to integrate sustainability into global operations. It shows how technological innovation and ethical leadership can transform industries that are traditionally seen as "hard to decarbonize." With growing consumer and regulatory pressure for sustainable business models, understanding freight decarbonization will be crucial for ethical decision-making and long-term competitiveness.

Speakers and Panelists:

- **Stefan GALLARD** – Marketing Director, Grain de Sail (a sustainable coffee and chocolate company using sailboats for transport).
- **Guilherme AZEVEDO** – Associate Professor, Organization Studies & Ethics, Audencia Business School.

The panelists shared their expertise on various aspects of sustainable transportation and discussed practical solutions to reduce environmental impact while maintaining efficiency and accessibility.

Theme 1: Over-Transportation and Invisible Costs

- Many products circle the globe multiple times before reaching consumers, highlighting inefficiencies in the current model.
- The low cost of transportation hides **externalities** like carbon emissions and ecosystem degradation, which are not reflected in market prices.

Theme 2: Rethinking What and How We Transport

- Transportation should be **reduced where possible**, focusing only on essential movement of goods.
- There's a call for **localized production** and **supply chain resilience**, which can decrease dependency on long-distance shipping.

Theme 3: Renewable Solutions and Technological Optimism

- Real-life examples include **solar-powered boats** in Peru and Egypt that replace fossil-fuel boats for essential services like student transport.
- Solutions for **decarbonizing coastal vessels and cargo ships** already exist – they just require support and implementation.

Theme 4: Nuclear Energy in Maritime Freight

- Discussed the controversial use of **small modular nuclear reactors (SMRs)** for powering large vessels.
- Although there are public perception and safety concerns, nuclear energy could be a viable **low-carbon power source**, especially when paired with synthetic fuel production (e.g., green methanol, ammonia).

Conclusion and Personal Reflections

This round table illuminated the urgent need to transform maritime freight systems. The technical capability to decarbonize already exists, but requires political will, ethical leadership, and revised cost accounting. As a future business leader, this session inspired reflection on how systemic changes, not just incremental improvements, are essential. It emphasized the importance of integrating ethics, innovation, and sustainability into supply chain decisions.

Use of AI

To write this report, I used ChatGPT to help transcribe key points from the discussion and structure my thoughts. However, the analysis and reflections are entirely based on the panelists' contributions and my personal interpretation.

Open Question

How can governments and private companies collaborate effectively to accelerate the global transition to sustainable freight without compromising economic efficiency or security?